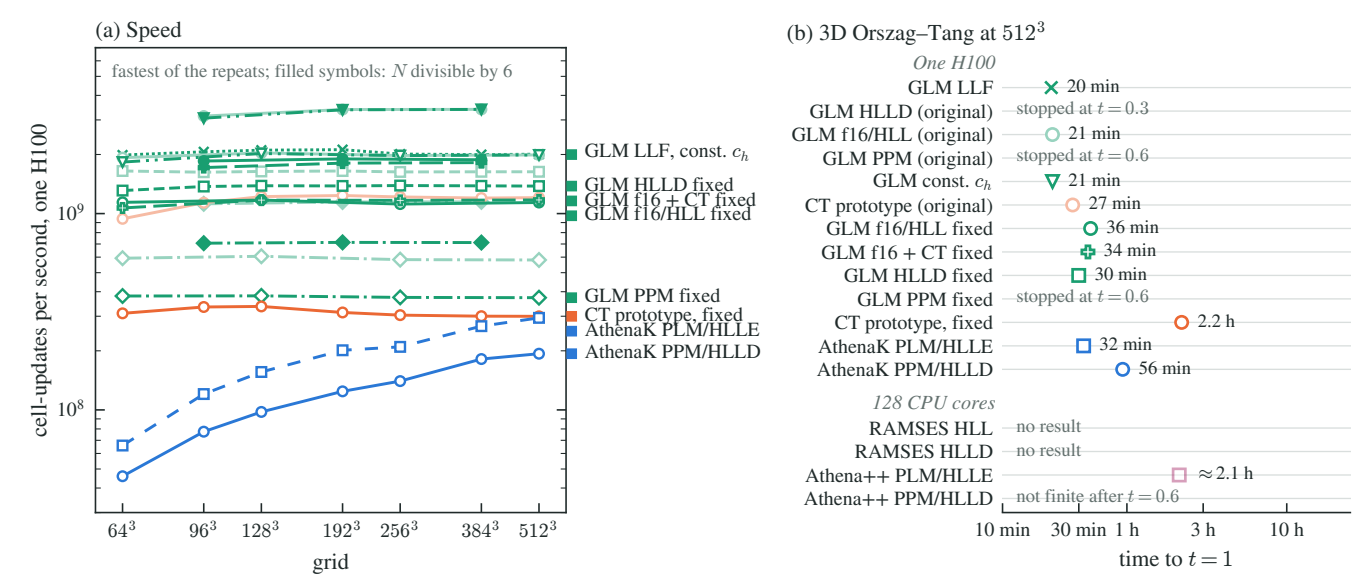
(a) Speed
fastest of the repeats; filled symbols: N divisible by 6
cell-updates per second, one H100
$10^9$
$10^8$
$64^3$
$96^3$
$128^3$
$192^3$
$256^3$
$384^3$
$512^3$
grid
GLM LLF, const. $c_h$
GLM HLLD fixed
GLM f16 + CT fixed
GLM f16/HLL fixed
GLM PPM fixed
CT prototype, fixed
AthenaK PLM/HLLE
AthenaK PPM/HLLD
(b) 3D Orszag–Tang at $512^3$
One H100
GLM LLF
20 min
GLM HLLD (original)
stopped at $t = 0.3$
GLM f16/HLL (original)
21 min
GLM PPM (original)
stopped at $t = 0.6$
GLM const. $c_h$
21 min
CT prototype (original)
27 min
GLM f16/HLL fixed
36 min
GLM f16 + CT fixed
34 min
GLM HLLD fixed
30 min
GLM PPM fixed
stopped at $t = 0.6$
CT prototype, fixed
2.2 h
AthenaK PLM/HLLE
32 min
AthenaK PPM/HLLD
56 min
128 CPU cores
RAMSES HLL
no result
RAMSES HLLD
no result
Athena++ PLM/HLLE
$\approx$2.1 h
Athena++ PPM/HLLD
not finite after $t = 0.6$
10 min
30 min
1 h
3 h
10 h
time to $t = 1$